12 The Exterior of the Watch

12.1 General remarks

The movement of a watch is a delicate object and must at all costs be protected. It must also be possible for it to be worn on the wrist, as a pendant, as a clip or carried in a pocket.

The display, hand-setting and correcting functions must be easy to use.

The inherent beauty of a watch means that its design may follow certain fashions, and extremely attractive models can be created. Each type of watch has its own particular image.

The **exterior** of a watch comprises all the parts added to the movement ***(13)*** which contribute to its final appearance, the protection of the movement, the way it is fixed in place and how it is used, etc.

These parts are *(fig. 12-1)* :

1 **the glass**
2 **the gaskets**
3 **the bezel**
4 **the case band**
5 **the crown**
6 **the correctors**
7 **the dial**
8 **the hands**
9 **the casing ring**
10 **the back cover**
11 **the lugs**
12 **the bracelet or strap**

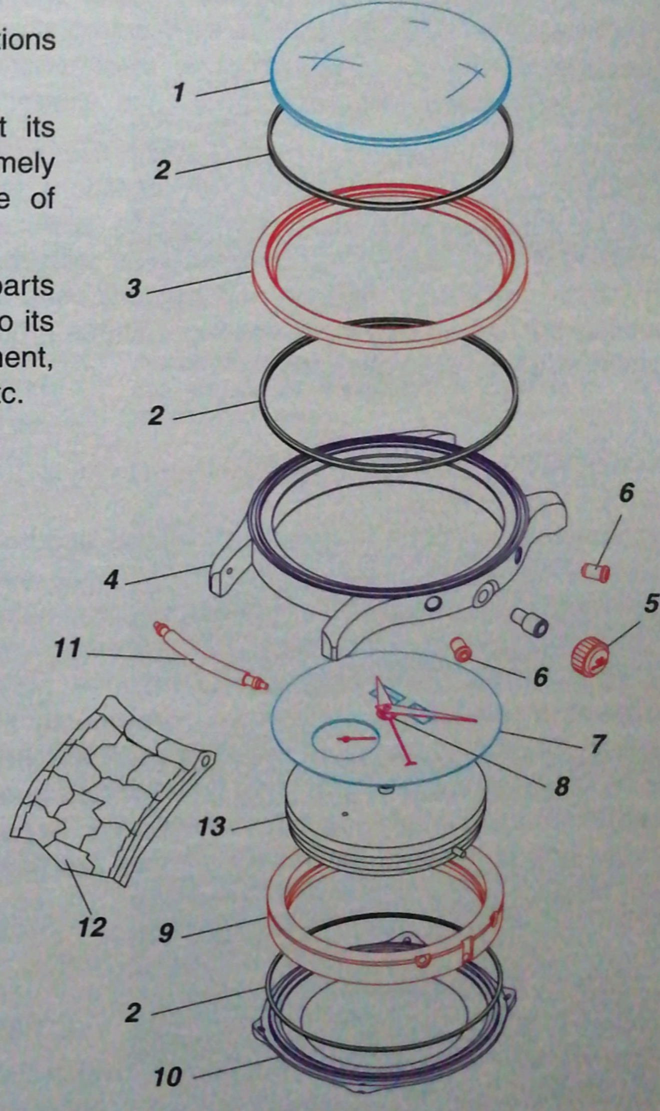


fig. 12-1

12.2 Health and the environment

Like other products, watches are subject to laws and regulations concerning health and environmental protection. Any country may refuse to allow the importation of watches which do not conform to its regulations. Although little consideration was given to such problems for many years, solutions now need to be found. Each manufacturer is obliged to respect the regulations.

fig. 12-2

Allergies among watch wearers

Certain metals, such as nickel or chrome for example, are included in alloys which are used to make cases and bracelets, for example stainless steel or certain alloys of precious metals. These metals are also used as a base for electroplating. They may sweat out of the alloys or electroplated deposit and come into contact with the wearer's skin.

Certain people (about 10% of the population) are allergic to these metals and develop a skin condition *(figs. 12-2 and 12-3)*. This also happens with the **synthetic materials**, **leather** and **cement** used in making cases and bracelets or straps. These products may also cause allergies.

fig. 12-3

Television	: 0.01 mSv
Nuclear power station	: 0.01 mSv
Watch with tritium hands and symbols	: 0.02 mSv
Flight (Zurich - New-York)	: 0.02 mSv
Radiation from the human body	: 0.3 mSv
Cosmic radiation	: 0.35 mSv
Earth's radiation	: 0.7 mSv
Radiation in the medical sector	: 1.2 mSv
Radon found in houses	: 1.5 mSv

fig. 12-4 : Annual dose of irradiation in Sieverts [Sv]; 1 SV = the absorbed dose of ionising radiation of 1 joule/kg.

Radioactive emissions

Radioactive luminous material used to be used on dials, hands and certain display discs. These materials included radium, tritium or prometheum salts. Although the levels of radiation are extremely low today, the manufacture of such luminous parts involves a risk for the workers who are directly involved. Certain countries do not allow such watches to be imported. It is important that the materials used correspond to current regulations and laws. In fact today, non-radioactive luminous materials are used more and more.

Disposal of batteries

As with all battery-operated appliances, watch batteries should be disposed of at special collection points so that the harmful substances they contain can be recycled *(fig. 12-5)*.

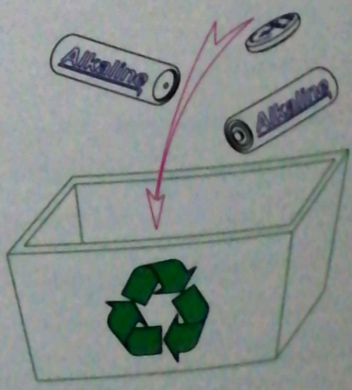


fig. 12-5

12.3 The limitations of a watch

Unlike many other products, a watch is subject to extreme chemical and physical limitations.

The exterior of the watch must be designed in such a way that the movement, and in particular its lubrication, cannot be damaged or its function impaired and that the exterior does not wear too quickly.

ISO and NIHS norms describe in detail the various ways of testing parts or assembled watches, of assessing the results in order to determine their level of resistance to various phenomena and of determining their quality.

fig. 12-6 : Stainless steel bracelet screw corroded through sweat (corrosion test using synthetic sweat).

fig. 12-7 : Knurled stainless steel bracelet pin corroded during a simulation test (combining the effects of corrosion, normal wear and mechanical wear).

12.3.1 Chemical limitations

A watch is permanently in contact with chemical agents which may be aggressive.

Deposits and dirt become encrusted in the nooks and crannies of the case and the bracelet, and cause corrosion and damage to the various external parts. These damaging chemical agents are listed below.

Atmospheric

Salty air at the sea-side or polluted air in built-up areas and industrial zones containing sulphurous or chlorinated acids may damage a watch.

fig. 12-8 : Chronograph push-piece screw corroded in actual wear by sweat and a salty environment.

Liquids

They may include sea-water, sweat or aggressive liquids in which a watch might be immersed.

Gases

Vapour emitted by the cement used in the manufacture of straps and presentation cases may damage the glass or the dial and impair lubrication.

fig. 12-9 : Dot of sulphurisation on a silvered dial resulting from contact with air containing H_2S and SO_2.

12.3.2 Physical limitations

Temperature

When a watch is in direct sunlight for any length of time its temperature may reach 70°C *(fig. 12-10)*. The temperature of a watch on the wrist normally remains quite constant at around 28°C. If it is not worn its temperature will be that of the surrounding air, which may fall below freezing point.

The materials used to make a watch, as well as the oil used to lubricate it, must be able to withstand such temperature changes.

fig. 12-10

Solar radiation

Almost any watch will be exposed to solar radiation, of which the ultraviolet rays may damage the varnish on the dial, the plastic used for the case and the strap or certain treated surfaces.

Shocks

When a watch is worn it suffers shocks which cause linear and angular acceleration or deceleration. The degree of linear acceleration in a wristwatch caused by abrupt movements and shocks *(fig. 12-11)* varies between around 10 g and 1000 g (1 g = the natural acceleration due to gravity $\cong 9.81\ m{\cdot}s^{-2}$). If, however, a watch with a heavy steel case falls on to a tiled floor the rate of deceleration may be as high as 8000 g.

fig. 12-11

Instruments such as the **striker** *(fig. 12-12)* exist for testing standard shocks. Others reproduce a series of accelerations at fixed intervals which are similar to those the watch will be subjected to during normal wear.

Magnetism

Watches worn under normal circumstances are exposed to an increasing number of magnetic fields created by electrical installations, television sets, various household appliances or even the magnetic seal or clip on a fridge door, a kitchen cupboard, a handbag or a briefcase.

Today watches must also be able to withstand such magnetic forces.

fig. 12-12

12.3.3 Water-resistance

The movement in a watch must be protected from dust, humidity and all liquids. The exteriors are designed to provide this protection, some more successfully than others. For aesthetical reasons, and in order to keep the size of the case within practical limits, protection against dust and humidity has often tended to be neglected, particularly in small and jewellery models.

fig. 12-13

Water pressure and water-resistance

If a watch, or any other object, is immersed in water the liquid exerts a pressure which varies with the depth of immersion.

Near the surface the pressure of the water will be close to air pressure. With every 10 m of increasing depth the water pressure rises by 1 bar (10^5 Pa). The water-resistant characteristics of a watch will depend on the use for which it is designed.

Standard water-resistance corresponds to normal use, i.e. humidity will not get into the watch through splashing, a shower or a even short swim.

Water-resistance in diving watches must be such that the movement remains dry at depths which a diver is likely to reach *(fig. 12-13)*. If a diver goes down to 100 m, for example, the watch must be able to resist water pressure of 10 bars.

fig. 12-14

Duration of water-resistance

A watch is water-resistant when it is checked before being sold. With time the effectiveness of the various gaskets (back cover, crown, glass, etc.) may be reduced and the watch will no longer be water-resistant. Each time a watch is opened it is important to check all the gaskets and replace any that are imperfect.

It is essential to check the water-resistance of a watch each time it has been opened and resealed.

There exist instruments to do this which measure mechanically the distortion of the case under pressure *(fig. 12-14)*.

Another type of instrument which involves columns of liquid can detect any leaks: the level of the liquid indicates the volume of air escaping from the watch *(fig. 12-15)*.

fig. 12-15 : Instrument designed to check 10 watches simultaneously.

12.4 Terminology of exteriors

1 Dial holes
2 Dial
3 Day and date apertures
4 Glass with ring
5 Bezel/case band
6 Movement fitting
7 Stem tube
8 Back-cover gasket
9 Dial feet
10 Screw-in back cover
11 Snap back cover
12 Crown
13 Push-pieces
14 Corrector
15 Symbols
16 Display cell
17 Gap between the lugs
18 Spring bar
19 Bracelet or strap

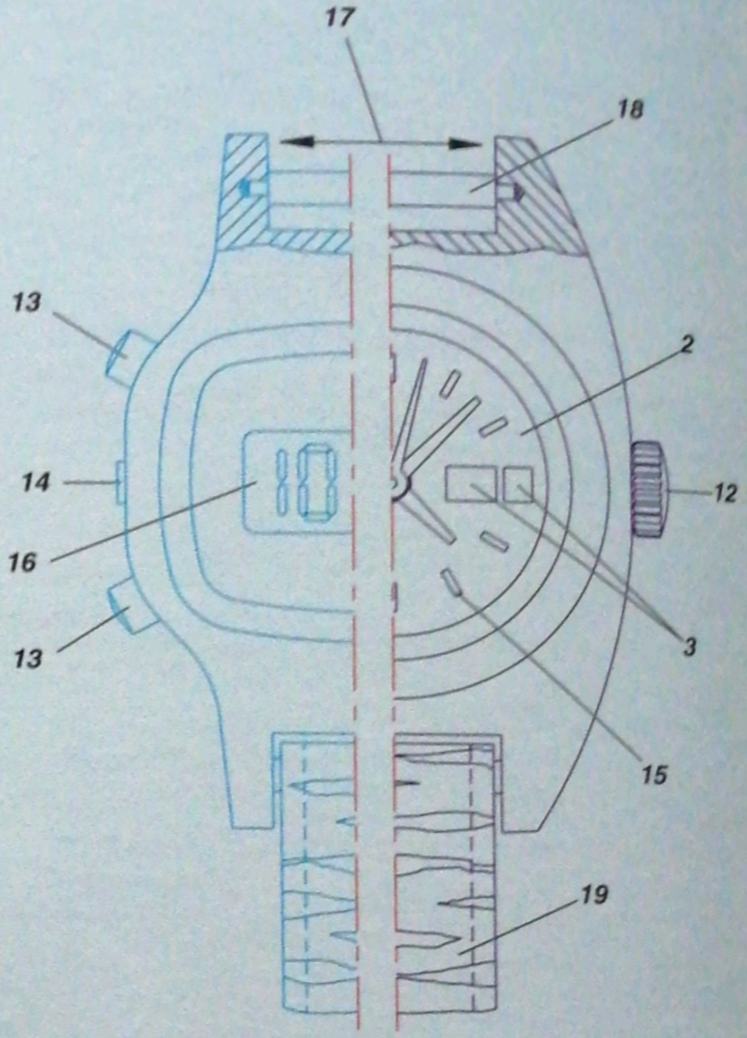


fig. 12-16 : Front view.

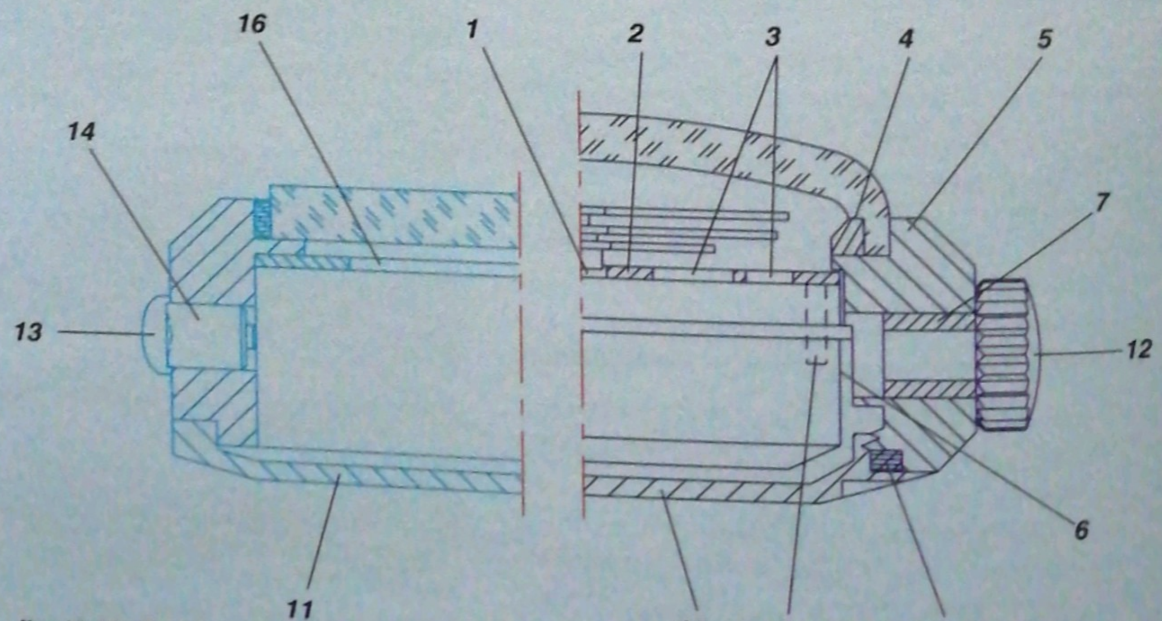


fig. 12-17 : Cross-section.

12.5 Time display

The principal function of a watch is to show the time, at a glance, if possible. A quantity of other information may also be shown. There are two ways of displaying this information.

12.5.1 Analogue display

This generally involves a principle established in the 13th century, which is of **hands** that rotate around a **dial**. At first only one hand was used *(fig. 12-18)* and from approximately 1700 on two hands of differing lengths were fitted; complicated watches may have several hands.

Analogue display *shows* the real time. With one glance the wearer can tell the actual time, but also see how much time has elapsed and how much is still to come. This is a rich and subtle concept. The hands do not even need a dial, their relative positions providing sufficient information.

Conventional interpretation of the position of watch hands is a language which transcends all forms of writing and other cultural barriers and is understood in every corner of the globe. It is for this reason that analogue display is still very popular today.

fig. 12-18

12.5.2 Alphanumerical display

Alphanumerical display has been popularised in recent years through mass-produced electronic watches, clocks and other instruments *(see Chap. 15)*.

This type of display has been used for a long time in mechanical timepieces, using the same principle as that of date-indicator discs. Discs or rollers bearing letters or numbers turn beneath apertures *(fig. 12-19)*.

fig. 12-19

12.6 The case

12.6.1 Fixing the movement to the case

It is essential that the movement is firmly fixed to the case.

The movement is positioned by its casing diameter and its height is determined by the casing flange. It is held in place by casing screws which are screwed into the plate or by clamps secured by casing screws.

If the casing diameter of the movement is smaller than the case, casing rings or enlarging rings may be used. The manufacturer may prefer to use a smaller movement or a smaller one may be fitted if a broken movement cannot be replaced.

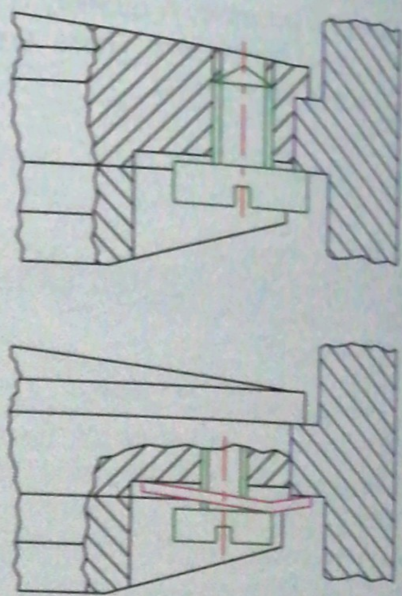


fig. 12-20 : Casing on the casing diameter.

Remark :

☞ an important dimension is the height of the winding stem in relation to the case; it should be identical for the movement and the case in order to ensure perfect coaxiality with the stem tube.

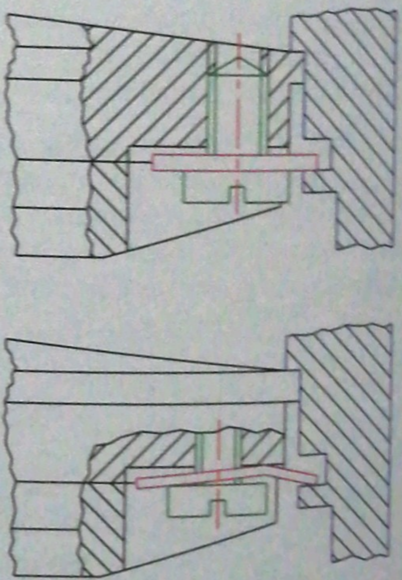


fig. 12-21 : Casing on the total diameter.

It is essential that there is a space between the movement and the case. This clearance may be only 0.2 mm for the fixed parts. It should be greater for the moving parts, the barrel, balance and oscillating weight in particular *(fig. 12-22)*, or for battery contacts. In certain cases a special cut-away may be made in the case.

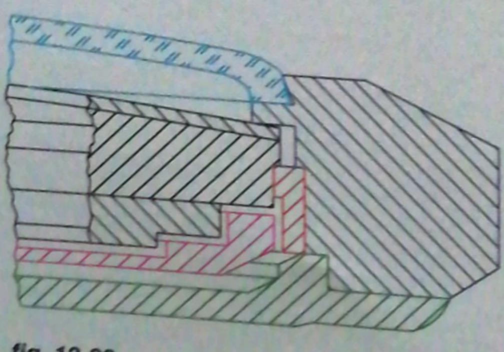


fig. 12-22

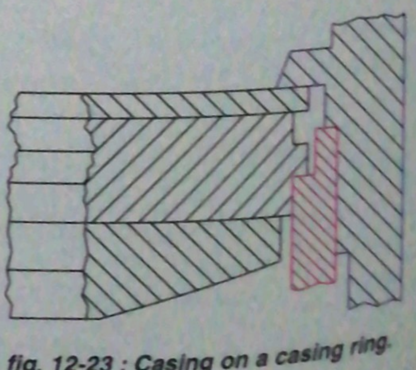


fig. 12-23 : Casing on a casing ring.

12.6.2 Different case designs

A watch case is made up of 3 distinct parts *(fig. 12-24)*: the **bezel** ***(1)***, the **case band** ***(2)*** and the **back cover** ***(3)***. These parts may form one piece, depending on the construction of the watch. The fourth indispensable part of the exterior is the **glass** ***(4)***.

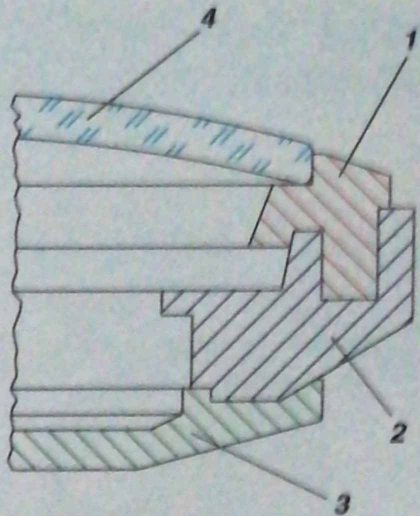


fig. 12-24

The "one-piece" case

In a *one-piece* case the bezel, the case band and the back cover are all one *(fig. 12-25)*.

The movement, along with the dial already fitted with the hands, can only be cased from the bezel side. The synthetic and flexible glass serves as the lid for the case.

The winding stem is in two pieces. The first section is part of the movement; the second, on to which the winding button is screwed, is fitted elastically on to the first, once the movement is in place *(fig. 12-25)*.

With the aid of a special clamp *(fig. 12-26)* the glass is squeezed together to reduce its diameter and placed in the circular groove around the inside of the bezel. Once the clamp is released it returns to its original shape and snaps into the groove, thus holding the movement in place *(fig. 12-27)*.

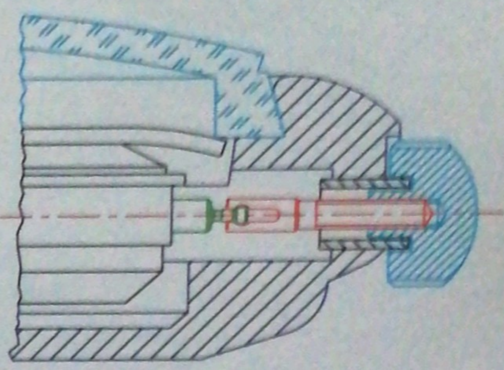


fig. 12-25

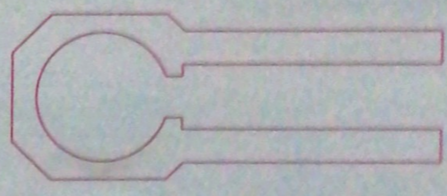


fig. 12-26

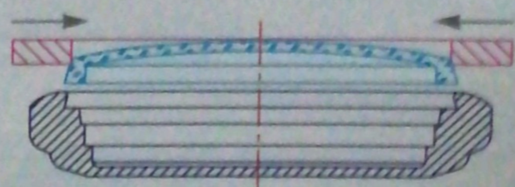


fig. 12-27

The "two-piece" case

In a *two-piece* case the bezel and the case band form one piece.

The glass is pressed or cemented into the groove in the bezel.

The back cover may be "snapped" into place using an 80° conical fitting, or it may be screwed on to the case-band *(fig. 12-28)*.

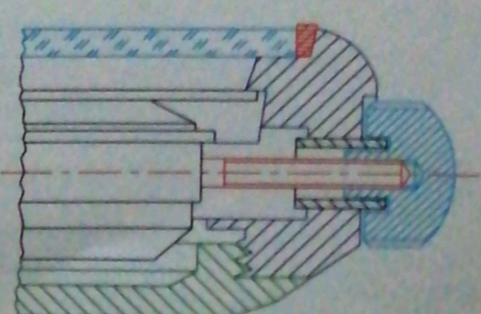


fig. 12-28

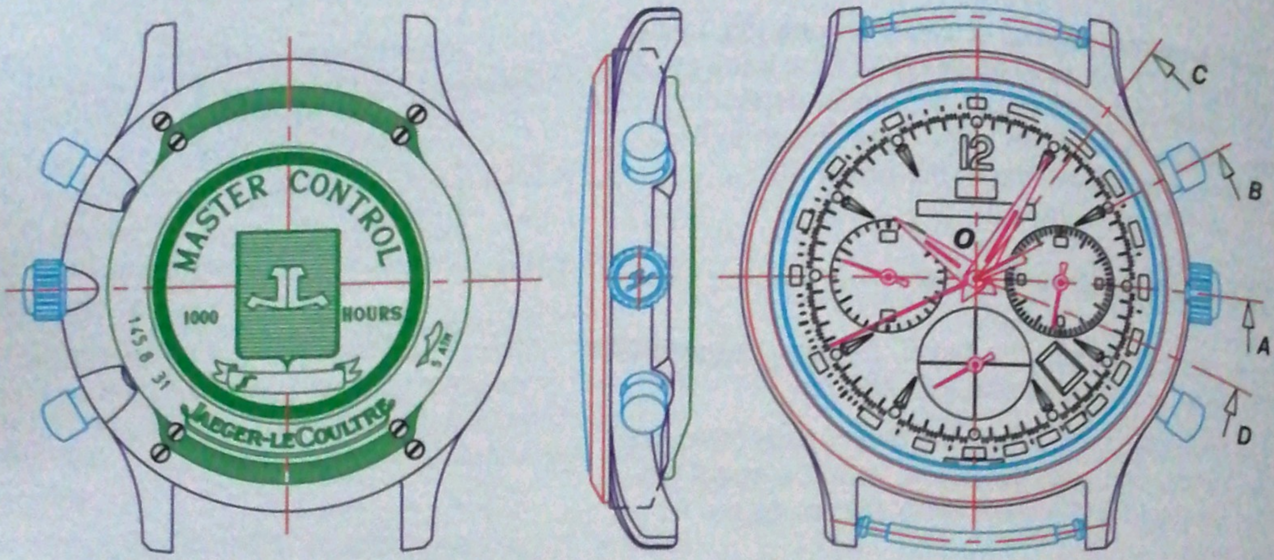
The "three-piece" case

In this type of case the bezel, the case band and the back cover are all separate *(see fig. 12-29)*.

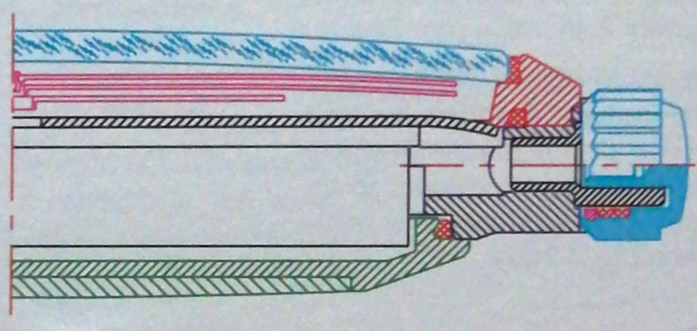
The bezel is snapped or screwed on to the case band.

The back cover is snapped or screwed into place.

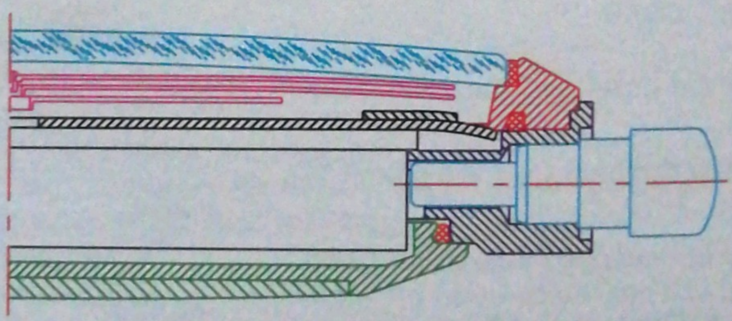
A sealed "three-piece" watch case *(fig. 12-29)*



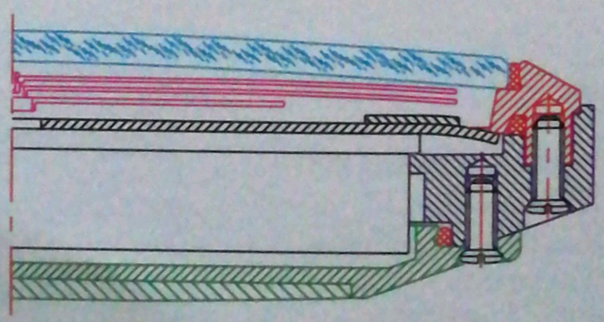
Cross-section O-A



Cross-section O-B



Cross-section O-C



Cross-section O-D

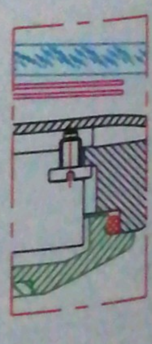


fig. 12-29 : Master Chronograph made by Jaeger-LeCoultre.

12.7 Winding crowns

The winding crown is screwed on to the winding stem. The wearer must be able to rotate it and to displace it axially in order to use the various functions.

To secure the crown it is also cemented on to the stem.

The winding stem fits into a hole in the side of the case. If this hole is not sealed, humidity, dust and fine fibres from clothing will get into the movement and may cause rust to build up which will stop the watch.

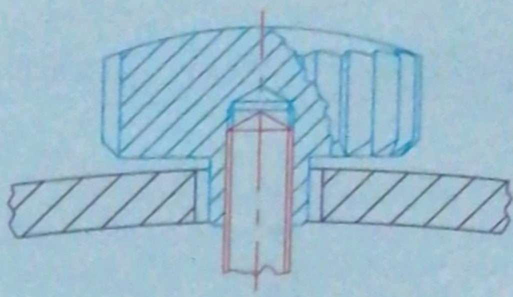


fig. 12-30 : Simple, unsealed crown.

The unsealed crown

The crown pipe rests in a hole in the case and fits more or less snugly in the case band. There must be a certain amount of play.

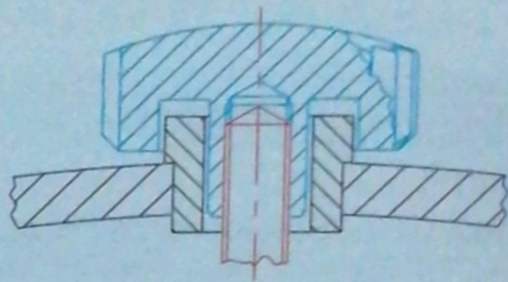


fig. 12-31 : Simple, hollow unsealed crown.

The sealed crown

The sealed crown turns on a tube pressed into the case. Water-resistance is ensured by an O ring.

Remark :

- winding crowns may be decorated with a cabochon stone or a decorative cap of a precious metal *(fig. 12-34)*.

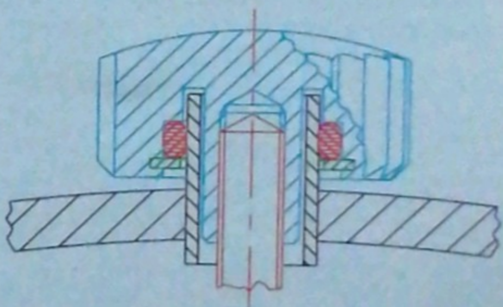


fig. 12-32 : Sealed crown and tube with gasket.

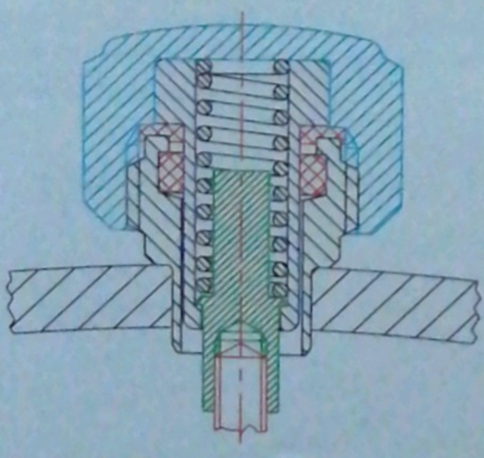


fig. 12-33 : Screwed, sealed crown.

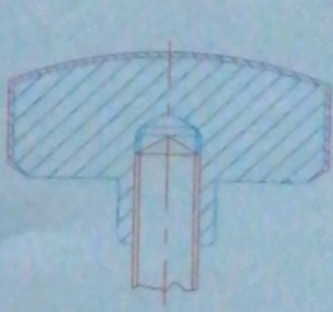


fig. 12-34

12.8 Push-pieces and correctors

12.8.1 Push-pieces

Push-pieces are buttons which project from the case band. Various functions can be operated by a push-piece. When pressure is withdrawn they return to their original position.

Push-pieces are used to control the functions of the chronograph, for example.

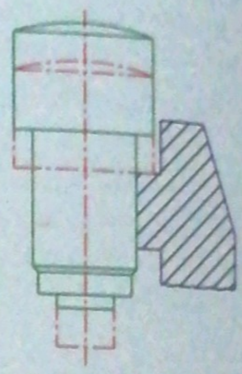


fig. 12-35 : Cylindrical push-piece.

12.8.2 Correctors

These are also buttons which are, however, countersunk into the case band. From an aesthetic point of view, this means that the case-band is free from protrusions.

Correctors may serve to alter different information shown on a complicated or an electronic watch, for example. Correctors can only be activated using a pointed instrument.

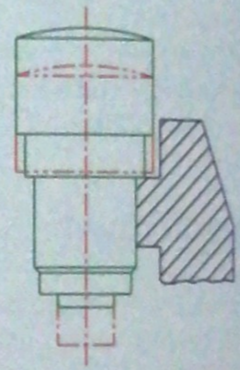


fig. 12-36 : Stepped push-piece.

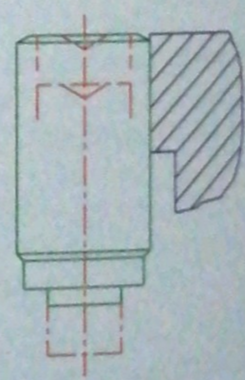


fig. 12-37 : Cylindrical corrector.

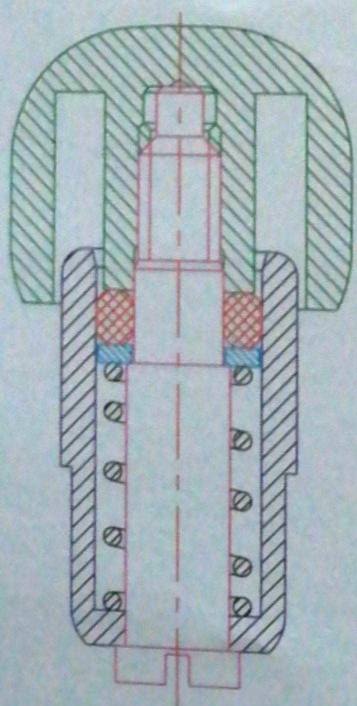


fig. 12-38 : Cross-section of a push-piece.

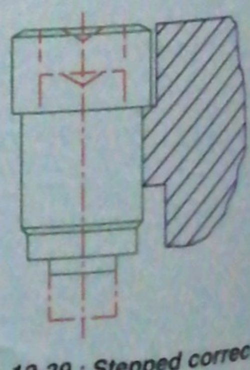


fig. 12-39 : Stepped corrector.

12.9 Watch glasses

12.9.1 Synthetic glasses

Since they are made of relatively soft material synthetic glasses are easily scratched. Most synthetic watch glasses are domed.

They are fitted and secured in the circular groove around the inside of the bezel through their own elasticity *(fig. 12-40)*. In some cases a metal ring runs around the inside of the glass to press it outwards against the groove, thus ensuring a good seal *(fig. 12-41)*.

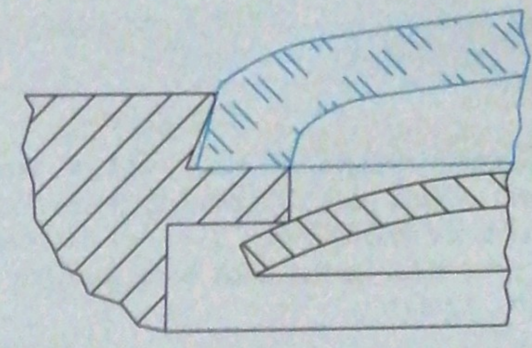


fig. 12-40

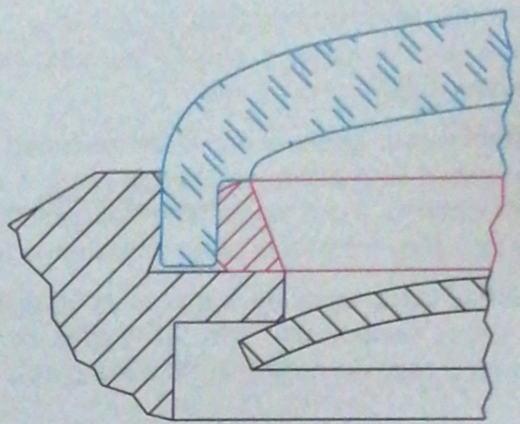


fig. 12-41

12.9.2 Natural or sapphire glasses

These watch glasses are not elastic since they are made out of a rigid material.

Those made out of natural glass may be scratched by very hard substances, while sapphire glasses can only be damaged by a diamond *(see 13.6.5)*. Sapphire glasses will remain in perfect condition for a long time. Any attempt to press them into a bezel groove would result in shattering, especially shaped glasses.

This means that they have to be cemented into the groove or fixed in place using a synthetic gasket *(fig. 12-42)*.

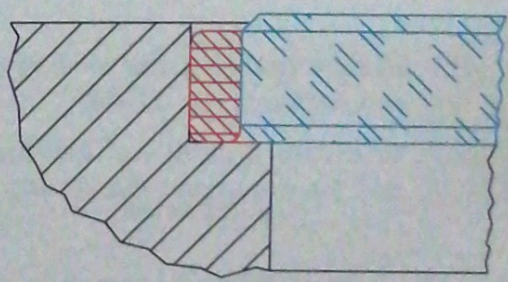


fig. 12-42

12.10 Gaskets and cement

12.10.1 Gaskets

Gaskets ensure a water-resistant seal between two moving parts, the winding crown and the crown tube, for example *(fig. 12-43)*, or between two fixed parts, such as the back cover and the case band *(fig. 12-44)*.

The gaskets can be in the form of O rings *(fig. 12-45)*, flat *(fig. 12-46)* or shaped *(fig. 12-47)*.

In principle, a gasket must always be elastic and should never be compressed if it is to fulfil its function. In this case it should be lubricated to allow some slide.

Back-cover gaskets are often designed to be compressed to a maximum. In this case, it is preferable to change them each time the watch is opened, since they may have been totally distorted.

Great care must be taken to ensure that no substance evaporates from the gasket or its lubricant which may damage the oil or the glass.

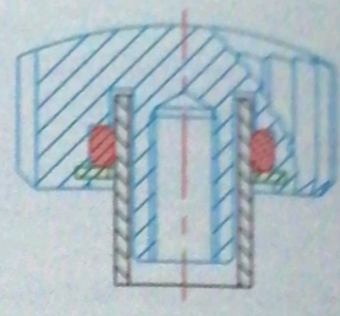


fig. 12-43

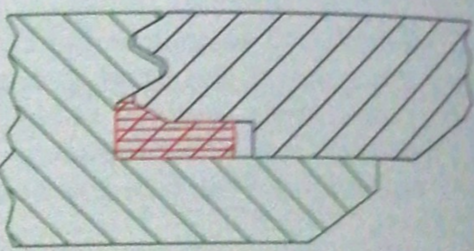


fig. 12-44

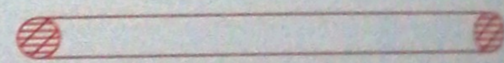


fig. 12-45

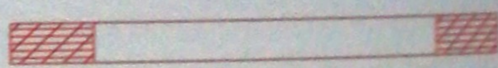


fig. 12-46

12.10.2 Cement

Cement is used for two purposes in watchmaking:

- **to fix a part:** e.g. the winding crown to the winding stem *(fig. 12-48)*;
- **to create a seal:** e.g. between the glass and the case band *(fig. 12-49)*.

Cement is in fact often used for both purposes at the same time.

The type of cement used is frequently **anaerobic**, i.e. it solidifies in the absence of air. This explains why jars of such cement are always only half-full (Loctite).

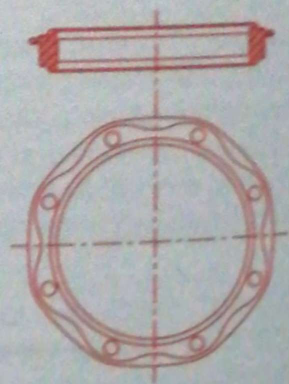


fig. 12-47

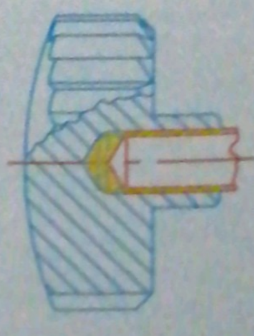


fig. 12-48

fig. 12-49

12.11 Dials

Dials are generally flat brass discs, bevelled or domed, which are fixed to the plate by dial feet.

The dials are either screwed to the plate via the dial feet *(fig. 12-50)*, held in place with synthetic inserts which are fixed to the plate and into which the dial feet are clipped *(fig. 12-51)* or held in place with clamps *(fig. 12-52)*.

Dials may have one or several round holes to contain the axes of the hands and apertures of different shapes for the date or phases of the moon for example *(fig. 12-56)*.

Minute, second and hour symbols, as well as the brand name, are normally transferred on to the dial by cliché-printing *(fig. 12-57)*, although silk-screen printing is occasionally used. The symbols give the appearance of being printed on the dial.

Hour symbols may also be applied to the dial and pinned in place *(fig. 12-53)*, cemented *(fig. 12-54)* or stamped into the dial *(fig. 12-55)*. Hour symbols are faceted, varnished and sometimes plated with a luminous substance.

The surface of the dial is finished by electroplating, varnishing or vapour deposition. In old watches the dials were made out of enamelled copper.

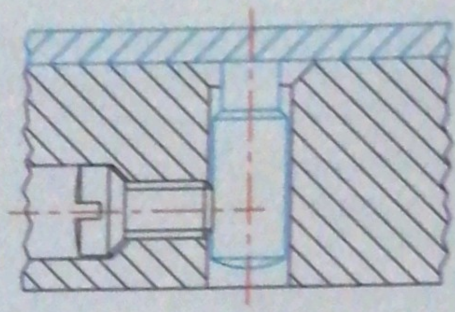


fig. 12-50

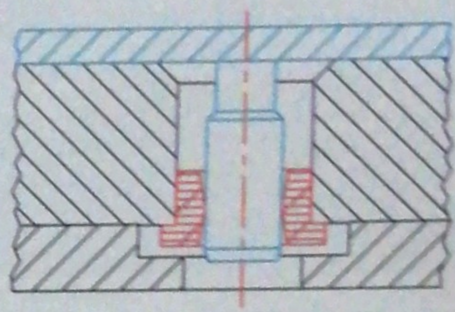


fig. 12-51

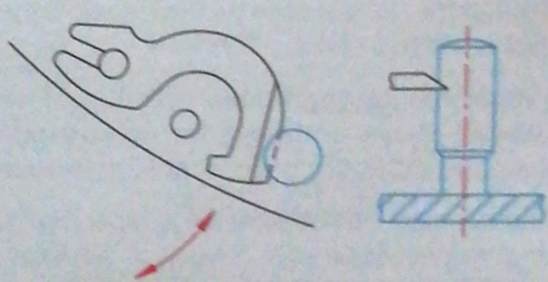


fig. 12-52

fig. 12-53

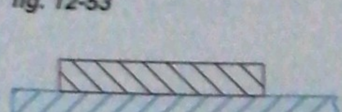


fig. 12-54

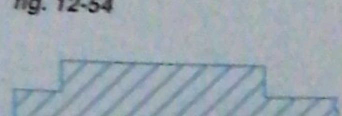


fig. 12-55

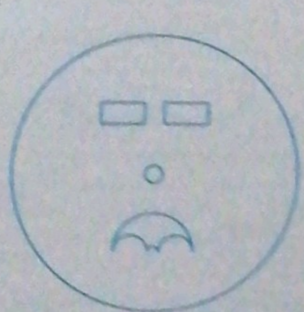


fig. 12-56

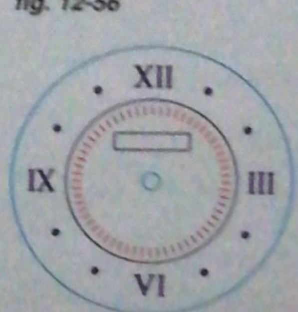


fig. 12-57

12.12 The hands

The hands of a watch are not simply indicators. Like the other exterior parts, the shape and appearance of the hands help to give a watch a certain style.

In technical timepieces, such as navigator's chronographs or stop watches, the hands are simple and functional.

In watches which are characterised by their style - modern or classical - the hands may be more complicated in their design. The different types of hands can be grouped into various categories: Alpha, Breguet, Dauphine, leaf, Louis XV, Louis XVI, pear, Roman or bâton, to mention but a few.

There are hundreds of different styles of hands.

The length of the hands also varies: the hour-hand is generally 20% shorter, but at the same time wider, than the minute-hand.

The main quality criteria are the type of surface and its durability, as well as fit, which must ensure that the hands remain firm on their axes if the watch suffers a violent shock.

Hands are made of brass, steel, gold, aluminium or special alloys and may be galvanised or painted, oxidised or, in the case of gold, left untreated.

Hands may be flat without a pipe *(fig. 12-63)*, have a stamped pipe *(fig. 12-64)* or be fixed to a separate pipe *(fig. 12-65)*.

fig. 12-58 : Alpha.

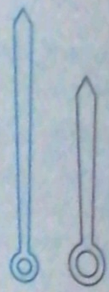


fig. 12-59 : Bâton.

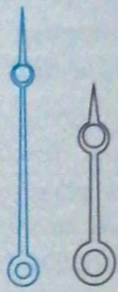


fig. 12-60 : Breguet.

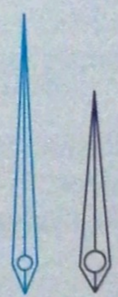


fig. 12-61 : Dauphine.

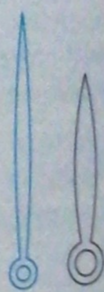


fig. 12-62 : Leaf.

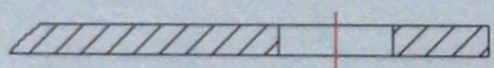


fig. 12-63

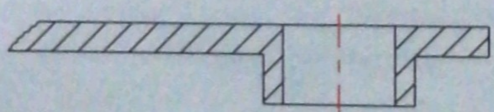


fig. 12-64

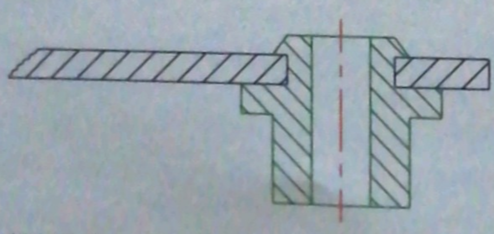


fig. 12-65

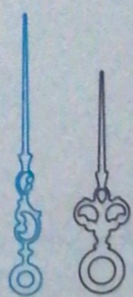


fig. 12-66 : Louis XV.

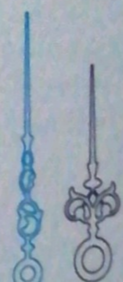


fig. 12-67 : Louis XVI.

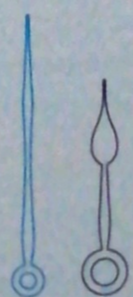


fig. 12-68 : Pear.

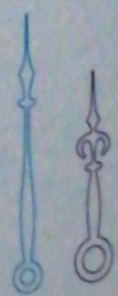


fig. 12-69 : Roman.

12.13 Materials used for exteriors

The material used to make a watch case must fulfil the following conditions: it must be corrosion and rust-proof, it must not be affected by contact with sweat and normal environmental conditions, and it must be hard-wearing. A watch case must remain in good condition for as long as possible *(fig. 12-70)*.

Watch cases are generally made out of electroplated brass, stainless steel, silver, standard gold of various colours or platinum.

Over the past few years watch designers have started to use other materials such as titanium, synthetic materials, ceramics, wood or stone, to mention but a few.

The bracelet is often made out of the same material as the case. Straps are normally made out of plastic or leather.

fig. 12-70

12.13.1 Precious metals

Precious metals recommended for use in jewellery include **gold**, **platinum**, **palladium** and **silver**.

Precious metals are never used in their pure form. They are alloyed with other metals to obtain a harder and more easily machinable material, as well as to alter their colour and lower the price.

Legal standards

The legal standard is the minimum amount of a given element expressed as thousandths of the total alloy (ISO).

Example : *(fig. 12-71)*

- **750 gold** (3/4 gold or 750 parts per 1000 → legal standard: 750);
- **18 carat gold** (3/4 gold or 18 parts of 24).

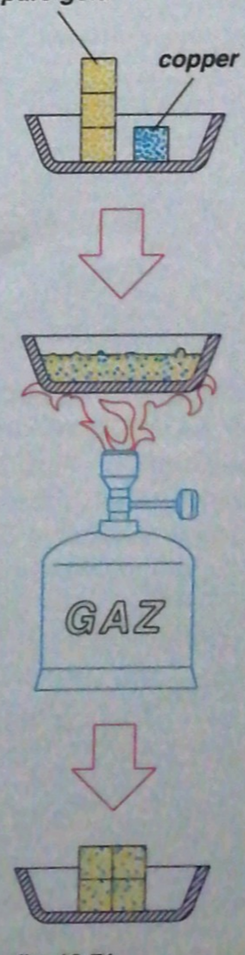


fig. 12-71

Legislation

Trade in precious metals has been the subject of legislation in all countries for a long time. This was all the more necessary when such metals were used as currency or as a currency standard.

As a rule, customs services control the import and export of precious metals as well as standards of alloys. Leglislation exists in every country and offenders are punished.

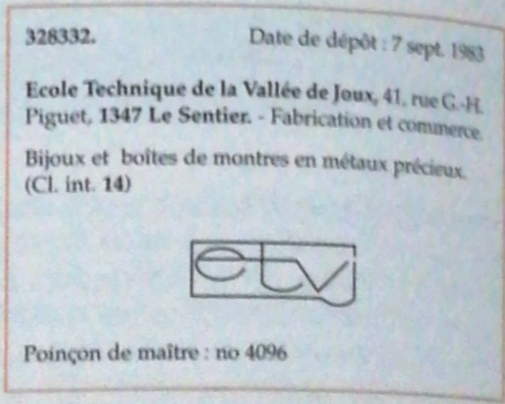
Switzerland has a federal office which deals with precious metals. It is a neutral body with no financial interests whatsoever. The main tasks of this office are to protect the public from fraud and to protect manufacturers from unfair competition.

Assay offices are to be found in various parts of the country, where **professional assayers** carry out the necessary analyses to:

- determine the amount of a precious metal in an alloy and to hallmark the piece if it comes up to standard;
- check that hallmarks are correct.

Every object made out of a precious metal should bear the following marks in close proximity:

- the **maker's mark**, which is registered with the authorities; this indicates who manufactured the object and guarantees the standard *(figs. 12-72 and 12-73)*;
- the **standard mark,** which indicates the legal standard in parts per thousand; this is used on watch cases for example *(fig. 12-74)*;
- the **hallmark**, which is stamped by the assay office. This is compulsory on all watch cases which contain a precious metal and optional on other objects. Since August 1, 1995 a hallmark in the form of *the head of a St. Bernard dog (fig. 12-75)* has been used in Switzerland for all metals and all standards. Old hallmarks applied before this date are still valid.


328332. Date de dépôt : 7 sept. 1983

Ecole Technique de la Vallée de Joux, 41, rue G.-H. Piguet, **1347 Le Sentier.** - Fabrication et commerce.

Bijoux et boîtes de montres en métaux précieux. (Cl. int. **14**)

Poinçon de maître : no 4096

fig. 12-72

fig. 12-73

fig. 12-74

fig. 12-75

Plating of common metals - German silver or brass for example - with precious metals is also subject to regulations.

Laws and norms

Each country has its own laws and regulations. National or international norms (ISO) also exist.

It is essential for all watchmakers to keep up-to-date with current legislation in this respect.

The customs and excise departments in each country can supply all the necessary information and documentation.

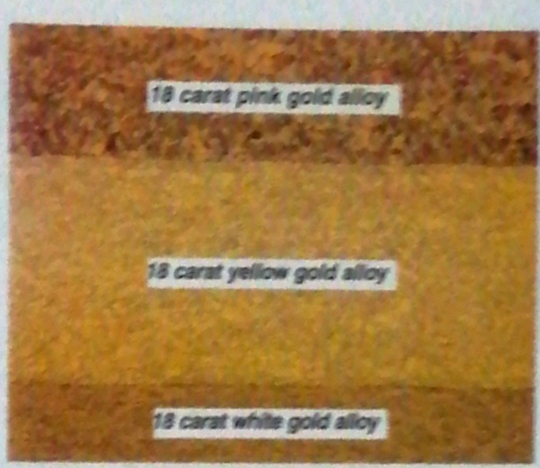


fig. 12-76 : Plating using three differents metals.

12.13.2 Stainless steel

Stainless steel is an ideal material for manufacturing watch cases. Thanks to its strength and durability it is a preferred metal for making resistant and hard-wearing exteriors.

Stainless steel is difficult to machine, however, and exteriors made in stainless steel using traditional methods (blanking, turning and milling) are not cheap. Cheaper cases can be produced using powdered-metal technology or **metal injection moulding (MIM)** *(see 12.13.4)*.

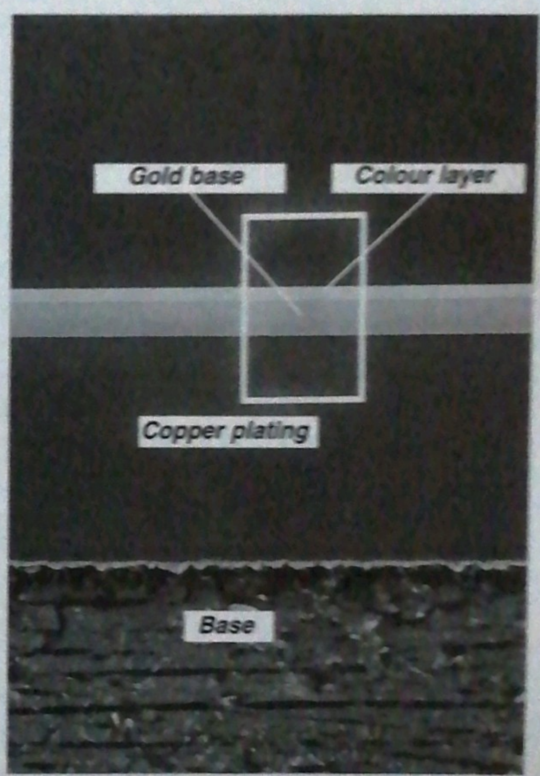


fig. 12-77 : Deposits on case band.

12.13.3 Brass

Brass is easy to machine and ideal for electroplating or plating with a thin layer of stainless steel or gold e.g. rolled gold.

If **MIM** is used the cost can be kept quite low.

Brass watch cases are cheap but their durability is limited. Once the protective layer has been worn away by friction or abrasion, brass is very quickly corroded by aggressive agents *(see 12.3.1)*.

fig. 12-78 : Deposits on case band.
1) Au/Pd, 2) Pd/Ni, 3) Ni/Co, 4) Au/Pd, 5) Ni/Co.

12.13.4 Ceramics

The English word comes from the Greek *keramos* meaning *potter's earth*. Since their ancient origins ceramics have evolved considerably and developments are still being made.

Ceramics are a composition of minerals and inorganic, non-metallic and crystalline substances, which are produced by all sorts of chemical processes between metallic elements (e.g. aluminium, magnesium, zirconium, titanium and tungsten) and non-metallic elements (carbon, nitrogen and oxygen).

The types of ceramics used for watch exteriors must be attractive, hard-wearing, scratch-proof, shock-resistant, chemically inert and not cause skin allergies. They can be grouped into four categories *(fig. 12-79)*.

Ceramics are **shaped** principally in one of two ways:

- **pressing:** whereby organic binding agents are added to the ceramic powder to enable simple blanks to be pressed out. With the increasingly complicated shapes and large quantities which are produced today, the general tendency is to use **ceramic injection moulding** (**CIM**).

  The same method is used for stainless steel and brass (**MIM**);

- **sintering:** this is a process whereby powdered ceramics are solidified at very high temperatures in the air or in a controlled atmosphere, with or without pressure. Solid phase diffusion, sometimes involving the formation of a film of liquid between the grains, produces a polycrystalline material with a vitreous phase or a film which binds the grains and a degree of porosity which should be as low as possible.

Zirconium-based ceramics are used a great deal in watchmaking *(fig. 12-80)*.

The basic powder is white and the final colour can be varied more or less at will by using different chemical processes.

- **oxides**, e.g. aluminium oxide (Al_2O_3) and zirconium oxide (ZrO_2);
- **tungsten and titanium carbides**, comprising tungsten carbide with cobalt or nickel as a binding agent (WC/Co or WC/Ni) and **cermets** which are titanium carbide with the same binding agents (TiC/Co or TiC/Ni);
- **hard coating** obtained using VPD or CVD such as titanium nitride (TiN) or using plasma treatment as in the case of diamonds;
- **specialities** with sapphire, a transparent monocrystalline from Al_2O_3 which is scratch-proof and used for watch glasses and back covers.

fig. 12-79

fig. 12-80 : Exterior parts (case bands and links) in black zirconium.

density	: 6000 $kg{\cdot}m^{-3}$
hardness	: 12 GPa
resistance to flexion	: 800-1000 MPa
no Ni or Co	: no allergic reaction

fig. 12-81 : Properties of 3Y-TZP black ceramic made from zirconium oxide (ZrO_2).

12.14 Bracelets and their attachment

Since the purpose of a bracelet or strap is to keep the watch in place on the wearer's wrist it should be hard-wearing and fixed firmly to the watch case.

The decorative role of the bracelet or strap is also important.

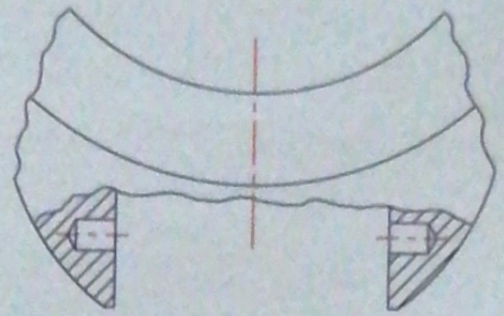


fig. 12-82 : Gap between lugs with blind holes.

12.14.1 The bracelet hard-soldered to the case

In general, if the bracelet is hard-soldered to the case it is made out of the same material.

If the bracelet is made out of a precious metal it must be of the same standard as the case.

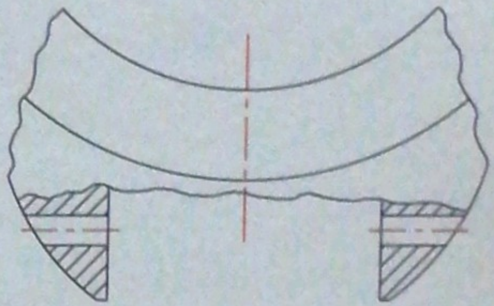


fig. 12-83 : Gap between lugs with through holes.

12.14.2 The removable bracelet

This may be made out of:

- any metal,
- a synthetic material,
- leather.

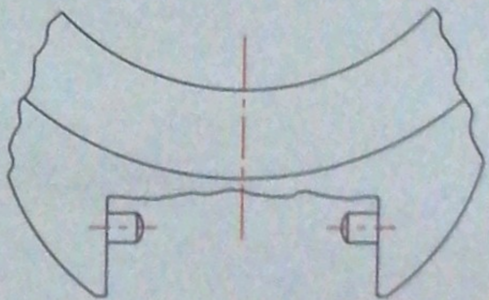


fig. 12-84 : Gap between lugs with pivots.

It is attached to the case with spring bars which are fixed between the lugs with pivots or holes.

The width of the bracelet or strap must be the same as the distance between the lugs. When a bracelet or strap is fitted special care should be taken to ensure that the spring bars are firmly in place.

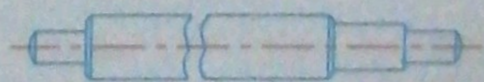


fig. 12-85 : Simple spring bar, with one moving end.

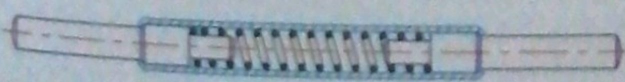


fig. 12-86 : Cross-section of spring bar with spring released.

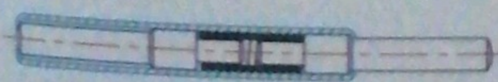


fig. 12-87 : Cross-section of spring bar with spring compressed.

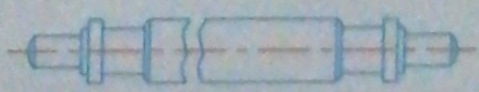


fig. 12-88 : Spring bar with a safety collar, and two moving ends.

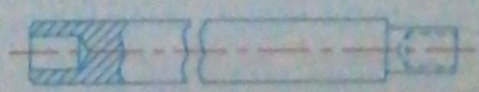


fig. 12-89 : Hollow spring bar.

Jean Rousseau, Geneva, c. 1660.

Roux and Bordier, Geneva, c. 1795.

Gounouilhou and François, Geneva, c. 1840.

Mido, Biel, c. 1930.

fig. 12-90 : Examples from the collection of the Geneva Watchmaking and Enamelling Museum (15, Rte de Malagnou - GE).